

PROJECT TOPIC: MUNICIPAL DRINKING WATER SUPPLY, PURIFICATION & DRAINAGE AUDIT

Topic Description: Comprehensive Engineering and Environmental Audit of Municipal Drinking Water Supply Infrastructure: Raw Water Sources, Water Treatment Plant (WTP) Purification Stages, Piped Distribution Networks, and Wastewater Drainage Outflows

The provision of safe, wholesome, and potable drinking water to urban and semi-urban populations is the most vital statutory public health responsibility executed by municipal corporations and urban local bodies. In the state of Maharashtra, urban agglomerations and regional townships—such as Pune, Mumbai, Nashik, Chhatrapati Sambhajinagar, and Nagpur—depend on vast, complex civil engineering hydraulic networks that convey raw water from distant surface storage reservoirs across dozens of kilometers to centralized municipal Water Treatment Plants (WTPs), distribute treated water through pressurized underground pipelines to household taps, and subsequently collect and evacuate millions of liters of used domestic wastewater.

A modern centralized municipal water supply system represents a sophisticated multi-stage technological chain spanning: **Raw Water Intake** from dams or rivers; **Physical and Chemical Purification** (Aeration, Coagulation with Alum, Flocculation, Rapid Sand Filtration, and Chemical Disinfection with Chlorine); **High-Pressure Pumping** to elevated Master Balancing Reservoirs (MBRs) and Elevated Storage Reservoirs (ESRs); and **Gravity-Fed Reticulation Networks** reaching residential neighborhoods. Equally critical—yet historically neglected—is the tail end of the urban water metabolism: the underground sewerage network that collects domestic greywater and blackwater for treatment before environmental discharge.

This project conducts a comprehensive field investigation and technical audit of the

2. Importance of the topic (विषयाचे महत्त्व)

Auditing municipal water supply, purification, and wastewater drainage systems is of paramount public health, ecological, and economic importance:

Crucial Reasons Why Water Supply Auditing Matters :-

- **Preventing Catastrophic Waterborne Epidemics:** Effective multi-stage water treatment (especially complete microbial chlorination) is the sole barrier protecting millions of citizens from lethal waterborne pathogens (Cholera, Typhoid, Hepatitis A and E, Rotavirus).
- **Detecting Massive Non-Revenue Water (NRW) Leakages:** Aging, corroded municipal distribution pipelines frequently lose between 30% to 45% of treated drinking water to subterranean pipeline fractures before it ever reaches consumer taps.
- **Uncovering Subterranean Cross-Contamination Risks:** When drinking water pipelines run in close physical proximity to unlined open sewage drains, intermittent negative pressure draws pathogen-rich sewage directly into potable drinking taps.
- **Halting Untreated Sewage Discharge into Natural Rivers:** Documenting the fate of municipal wastewater drainage exposes the massive deficit in municipal Sewage Treatment Plants (STPs), holding civic authorities accountable for river pollution.

A transparent, scientifically audited municipal water supply network is the primary foundation of civilized urban existence.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This drinking water supply and drainage audit was formulated around structured engineering and environmental objectives:

Structured Project Objectives :-

- **Raw Water Source & Conveyance Mapping:** To identify the primary surface reservoir feeding our locality, recording raw water storage capacity, intake elevation, and pipeline transmission distance.
- **Water Treatment Plant (WTP) Process Audit:** To visit and document the sequential purification stages at the municipal WTP: Aeration, Alum Flash Mixing, Clariflocculation, Rapid Sand Filtration, and Chlorination.
- **Distribution Network & Zoning Evaluation:** To inspect the Elevated Storage Reservoir (ESR) serving our ward, documenting daily supply hours, per-capita supply volume (LPCD), and distribution pressures.
- **Wastewater Drainage & Outflow Tracking:** To trace the physical flow of domestic greywater and blackwater from household drains into municipal sewers, evaluating treatment efficiency at the municipal STP versus untreated river discharge.
- **Formulating Public Health Recommendations:** To synthesize actionable guidelines for domestic water storage hygiene and community water conservation.

The study provides an objective empirical evaluation of urban municipal water infrastructure.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A structured field inspection, technical process auditing, and municipal engineering interview methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Municipal Permission & Administrative Data Collection:** Secured academic permission from the Municipal Corporation Water Supply & Sewerage Department; collected official engineering schematics, ward population maps, and daily supply quotas.
- **Phase 2 – On-Site Water Treatment Plant (WTP) Audit:** Conducted a guided 3-hour technical inspection of the municipal WTP (capacity: 120 MLD), logging chemical dosages (Alum, Chlorine gas), filtration rates, and backwashing cycles with the plant chemist.
- **Phase 3 – Neighborhood Distribution Survey:** Monitored tap water supply hours and pressure across 15 households in our ward over 7 consecutive days; tested residual chlorine at household taps using an orthotolidine (OT) color comparator.
- **Phase 4 – Wastewater Outfall & Drainage Inspection:** Traced the primary municipal trunk sewer lines leaving our residential ward to inspect the terminal sewage pumping station and the municipal Sewage Treatment Plant (STP) outfall.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 2 (Water Pollution) and Chapter 5 (Water Security: Water Supply and Treatment) of the HSC Standard 12 EVS curriculum.

All recorded engineering capacities reflect verified municipal administrative records and on-site testing data.

5. Observations (निरीक्षणे)

Technical field inspection of the municipal water purification and distribution infrastructure revealed detailed engineering parameters and critical systemic challenges:

5.1 Municipal Water Supply System Flowsheet: From Reservoir Source to Household Tap

System Stage	Infrastructure Specifications & Technical Operations	Performance Evaluation & Observed Findings
1. Raw Water Source & Intake	- Source: Khadakwasla / Regional Dam Reservoir (Distance: 18 km). - Raw water conveyed via closed mild-steel gravity conduits (1,800 mm diameter) to WTP.	Gravity flow minimizes pumping electrical costs; raw water exhibits low baseline turbidity (<15 NTU) during dry seasons, spiking to 120 NTU during monsoon floods.
2. Water Treatment Plant (WTP) Purification	- Cascade Aerator (removes volatile gases/odors) ightarrow - Flash Mixer (Alum dosage: 18 to 25 mg/L) ightarrow - Clariflocculator (settles chemical floc) ightarrow - Rapid Gravity Sand Filter (sand + gravel media) ightarrow - Post-Chlorination (Chlorine gas: 2.2 mg/L).	Highly effective physical-chemical purification: filtered water achieves pristine clarity (Turbidity <0.8 NTU); post-chlorination ensures 1.5 mg/L free residual chlorine at plant exit, eliminating 99.9% of coliform bacteria.
3. Storage & Piped Distribution	- Pure water pumped to 2 Elevated Storage Reservoirs (ESR capacity: 25 Lakh Litres each). - Intermittent gravity supply: 2 hours daily (6:00 AM to 8:00 AM).	Average per-capita supply: 140 Litres/capita/day (LPCD) ; residual chlorine at tail-end household taps tested at 0.3 mg/L (satisfactory, prevents secondary bacterial growth).
	5 Locality generates ~110 LPCD of wastewater (80% of supply)	Major Systemic Deficit: Locality trunk sewer operates well, but the

6. Analysis of data (माहितीचे विश्लेषण)

Hydrological and environmental analysis of the urban water supply and drainage metabolism exposes critical systemic realities:

Core Analytical Inferences :-

- **The Rigor of Multi-Barrier Purification:** Inspection of the municipal WTP confirms the absolute necessity of the multi-barrier purification principle. Coagulation with alum destabilizes microscopic negatively charged clay colloids, allowing them to agglomerate into heavy flocs that settle in clarifiers. Rapid sand filtration mechanically intercepts 98% of remaining suspended solids, while post-chlorination provides vital residual disinfection that persists throughout miles of underground distribution pipelines.
- **The Inherent Dangers of Intermittent Water Supply:** Unlike developed world 24x7 pressurized systems, intermittent water supply (2 hours daily) is hazardous. When pipes are depressurized for 22 hours, negative hydraulic pressure sucks surrounding contaminated groundwater, soil bacteria, and sewer leaks through hairline pipe fractures. This explains why household taps frequently run turbid, brownish water during the first 5 minutes of supply. Transitioning to 24x7 pressurized supply is an urgent public health imperative.
- **The Severed Water-Waste Loop:** While the municipal corporation invests hundreds of crores in treating raw reservoir water to pristine potable standards, it neglects wastewater. Millions of liters of purified water are used once to flush toilets and wash clothes, then dumped as raw sewage into natural rivers. Circular urban water management requires treating 100% of municipal wastewater to tertiary standards for industrial reuse and urban toilet flushing.
- **Household Storage Contamination:** Testing revealed that while water arriving from the municipal main was microbiologically sterile, water stored inside uncleaned household underground sumps and rooftop syntax tanks showed high bacterial contamination, proving that 6 unregulated storage hygiene is a critical weak link.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Auditing the municipal water supply system reveals an impressive civil engineering feat in raw water conveyance and centralized purification, but highlights alarming vulnerabilities in distribution pipe leakages, intermittent supply back-siphonage, and inadequate sewage treatment. Achieving true urban water sustainability requires upgrading distribution pipes, modernizing STPs to achieve 100% sewage recycling, and practicing domestic water conservation.

Action Checklist for Drinking Water Safety & Waste Reduction :-

- ❑ Always discard or use for plants the first 2 minutes of tap water when municipal supply resumes to flush out stagnant pipe sediment.
- ❑ Clean and disinfect domestic underground sumps and overhead rooftop water storage tanks at least once every 6 months.
- ❑ Boil drinking water or utilize certified UV/UF multi-stage filtration to eliminate potential secondary distribution pathogens.
- ❑ Promptly report roadside municipal pipeline bursts and visible water leaks to the local ward engineering office.
- ❑ Segregate domestic greywater (washing machine and bath water) for garden irrigation rather than sending it to sewers.

8. Bibliography / References (संदर्भ सूची)

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